When Is Video Enough? Predicting Deformable Object Responses to New Interactions

Anonymous CVPR submission

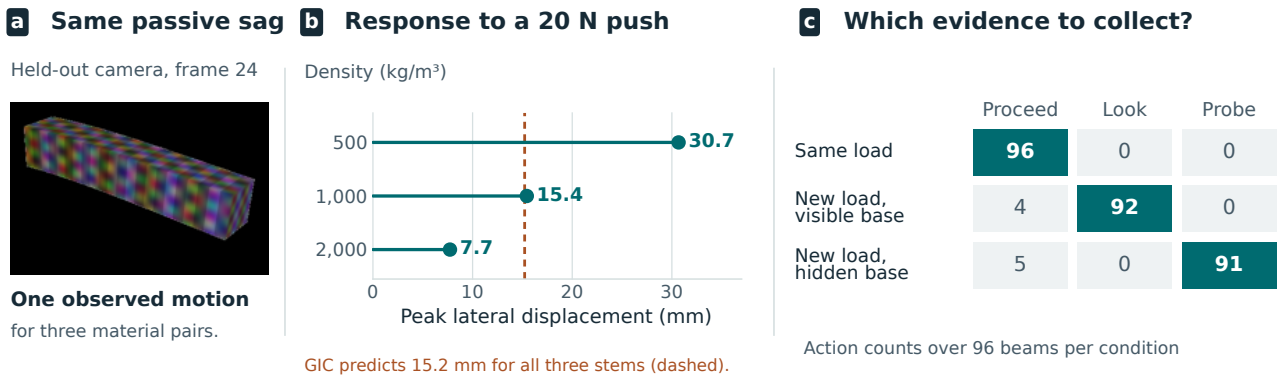


Figure 1. **The next interaction determines what the robot needs to know.** In the left and centre panels, three simulated stems have the same sag, yet move by 7.7–30.7 mm under a 20 N lateral push. GIC fitted to the passive images predicts 15.2 mm in all three. The right panel shows how, in a separate 96-case beam study, the initial tip observation can justify proceeding, observing the support, or probing, depending on the target and the accessible view.

Abstract

Three simulated elastic stems produce indistinguishable videos as they sag under gravity, yet deflect by 7.7, 15.4, and 30.7 mm under the same lateral force. Whether visual observations suffice to predict an object's motion therefore depends on the interaction to be predicted. We introduce a method that estimates this task-specific uncertainty and uses it to decide whether to proceed, acquire another view, or apply a calibrated probe. Given an object's rest geometry and surface motion observed from calibrated cameras, we infer a distribution over material properties, support conditions, and loads consistent with the observations. When displacement depends linearly on unknown load amplitudes, we marginalize those amplitudes analytically to preserve the dependence between inferred force and stiffness. We simulate the proposed interaction under each plausible physical model and estimate the probability that the prediction error exceeds a specified tolerance. We evaluate held-camera rendering alongside physical prediction and specify a common RGB benchmark for stem bending, off-centre loading of a heterogeneous block, and striker impact. On a held-out diagonal push, one calibration interaction reduces 3D displacement error from 5.3–12.1 mm to 0.65–2.75 mm across three stems using RGB-D observations. In a separate 96-case study, the system proceeds for a repeated load, generally selects another view when support motion is visible, and probes when the support is concealed.

1. Introduction

Consider a robot observing a flexible stem as it sags. A light, soft stem and a heavy, stiff stem can trace the same path in every camera. If the robot wants to repeat the release, the ambiguity need not affect its prediction. If it intends to push sideways with a known force, the two stems can bend by very different amounts. Better pictures of the same gravity-driven trajectory cannot settle this particular question. One force measurement with its corresponding displacement can.

A physical twin must therefore answer two questions about a proposed action. What response does the available evidence predict? How much would an additional observation improve that specific prediction? Rendering and simulation systems can recover striking visual and physical detail [3, 6, 7, 12]. Yet a small error on observed pixels leaves open whether the next error in deformation, or in transferred momentum, will be tolerable. Conversely, a poorly identified material parameter can have little effect on the response of interest.

We use a distribution of physical explanations for the observed motion. For each explanation we predict the same target response. Predictions that agree within the required tolerance support proceeding even when their material parameters differ. When they disagree, we evaluate what another camera measurement or a known probe could resolve before paying its cost. In the spatial identification model, unknown forces have their own locations and time histories. Integrating out uncertain force amplitudes avoids committing to a single load that happens to explain the observed displacement.

We make three contributions. First, we connect uncertainty over material, supports, and applied forces to the error of a specified interaction, including an explicit criterion for when visually equivalent explanations are incompatible with the same tolerable prediction. Second, we express another view, a calibrated probe, and proceeding without more evidence in terms of downstream predictive loss. Third, we test distinct parts of this procedure in controlled studies. Sparse 3D surface measurements test spatial inference, a compliant-beam cohort tests the decision to measure, and held-camera video and prospective stem forecasts test the relationship between visual fit and physical response. We report which parts of the method each experiment exercises.

2. Related Work

From images to a physical model. PAC-NeRF estimates geometry, appearance, and physical properties from multi-view video using a material-point simulation [7]. GIC uses time-dependent Gaussian reconstructions, surface data, and rendered masks to identify material parameters [3]. Phys-Gaussian transports Gaussian primitives with continuum motion [12], while Spring-Gaus and PhysTwin model dynamic objects with springs and Gaussian appearance [6, 14]. ReconPhys estimates physical attributes and appearance from a monocular video [11]. M-PhyGs identifies heterogeneous stiffness and density from multiview object interactions [10]. We use known rest geometry to study which predictions remain ambiguous even when geometric reconstruction is accurate.

Forces, probes and uncertainty. Latent-force models combine Gaussian process priors with physical observations [1, 8, 9]. Goal-oriented Bayesian design selects measurements for a downstream quantity [2]. KnockGS estimates effective elasticity and density from a known probe and freezes them to predict a second interaction [5]. PhysCoRe infers spatial elasticity and uses local confidence to guide further robot interaction [13]. Our question concerns the error of a particular prediction: an uncertain region can be irrelevant for one target and decisive for another. We compare both non-contact observation and probing against the option to proceed. The equal-evidence point-estimate and Gaussian-risk controls below test how much of the result requires a full posterior.

3. Predicting an Interaction from Partial Evidence

Let D contain the available motion observations and known input commands. A physical explanation z consists of the relevant material, boundary, and loading variables. A forward model produces the visible observations $g(z)$ and the response $q(z, a_\star)$ to a specified future interaction $a_\star$. The target may be a trajectory of surface points or an impulse delivered during contact. The command is available at prediction time. Its realized response is withheld.

When does an ambiguity matter? Suppose two explanations z_1, z_2 produce the same noiseless observations. Every estimator using those observations and the same target command must give them the same point prediction $\hat{q}$. For a response metric d the triangle inequality gives

$$\max_{i=1,2} d\big(q(z_i, a_\star), \hat{q}\big) \;\geq\; \tfrac{1}{2} d\big(q(z_1, a_\star), q(z_2, a_\star)\big). \quad (1)$$

If the two responses differ by more than 2δ, no single estimate can meet tolerance δ in both worlds. In Fig. 1, even a perfect passive fit must miss at least one extreme stem by about 11.5 mm in peak lateral deflection. Changing the task to a repeated gravity release collapses this separation. This bound is a worst-case statement. Estimating the probability of a threshold violation additionally requires a distribution over the plausible explanations.

Prediction and a decision to measure. We assign posterior weights to the physical hypotheses using D. Under squared error the point forecast is the posterior mean, $\hat{q} = \mathbb{E}[q(z, a_\star) \mid D]$. For any specified point forecast, its estimated task risk is

$$r(D, a_\star) = \Pr\big[d\big(q(z, a_\star), \hat{q}(D, a_\star)\big) > \delta \mid D\big]. \quad (2)$$

For a finite posterior this is the sum of the weights of hypotheses outside the tolerance. It can be large even for an excellent image fit. Its calibration depends on the forward models and sensor noise represented in the posterior.

Let s be a candidate measurement, y_s its possible outcome and c_s its cost in units of threshold failure. We choose the lowest-cost option from

$$J(\text{proceed}) = r(D, a_\star), \quad (3)$$

$$J(s) = c_s + \mathbb{E}_{y_s|D,s}\big[r(D \cup \{y_s\}, a_\star)\big]. \quad (4)$$

For each hypothetical observation we update *both* the weights and the point prediction. The expectation uses possible measurement outcomes available from the model, never

the true future response. The beam experiment uses threshold loss and different costs for looking and pushing. The spatial field experiment evaluates an analogous squared-error design for a fixed number of scalar measurements.

4. Physical Explanations and Observation Models

Spatial material, support and load. Figure 2 shows the unknown-load case. In the joint-field study the object is a known, small-strain elastic volume with displacement u_t from its rest configuration. Its positive Young's modulus is $E(X) = \exp(m_0 + m_1\chi(X))$ at material coordinate X, where χ is the volume-centred difference of two normalized Gaussian features. Two additional Gaussian surface footprints describe a compliant support and an applied traction. Each surface footprint integrates to one over its surface. Its position describes *where* the interaction acts, while each traction amplitude gives a total force in newtons. The support has stiffness k_b and energy $\frac{1}{2}k_b \int_{\Gamma_b} w_b(X)\|u\|^2 dA$. Two signed traction modes point downward on the top and laterally on a side. They share an inferred longitudinal centre c_f. The field hypothesis is $\theta = (m_0, m_1, c_b, \log k_b, c_f)$.

For each candidate, the tetrahedral stiffness operator K_θ and the unit-load map B_θ give

$$K_\theta u_t = B_\theta a_t + f_t^{\text{known}}, \qquad y = H_\theta a + d_\theta + \epsilon. \tag{5}$$

Here a stacks two force amplitudes at each of eight times. Solves for unit loads and projection into calibrated cameras define H_θ; d_θ is the known-load contribution. Scaling material stiffness, support stiffness and an unknown force by the same factor preserves the unknown-load displacement. An independently calibrated force breaks this symmetry. A Gaussian support footprint locates a mechanical interaction; a rendering Gaussian carries appearance. Only the former enters the stiffness operator.

Marginalizing unknown load histories. Conditional on θ, let $a \sim \mathcal{N}(0, \Lambda)$ with temporally correlated amplitudes and $\epsilon \sim \mathcal{N}(0, R)$. Gaussian integration gives

$$C_\theta = R + H_\theta \Lambda H_\theta^\top, \tag{6}$$

$$p(y \mid \theta) = \mathcal{N}(y; d_\theta, C_\theta), \tag{7}$$

$$\mu_\theta = \Lambda H_\theta^\top C_\theta^{-1}(y - d_\theta), \tag{8}$$

$$S_\theta = \Lambda - \Lambda H_\theta^\top C_\theta^{-1} H_\theta \Lambda. \tag{9}$$

The last two quantities describe the conditional force posterior. The likelihood includes $\log|C_\theta|$ as well as the residual. Thus a stiff body can fit the same measured displacement as a soft body while requiring a different force distribution. Predicting a hidden displacement under the original load uses the corresponding force conditional; a known reset load requires a forward solve for each material/support hypothesis. In both cases we integrate over the posterior weights. The unknown-force calculation applies where the observation model is affine in force amplitudes. Contact and impact forecasts below use known commands and direct rollouts instead.

An enumerated 5^5 dictionary provides exact weights for its 3,125 spatial hypotheses. An independent implementation uses adaptive tempered sequential Monte Carlo in the five continuous coordinates [4], retaining the Gaussian force conditional. The continuous study replays the finite planner's actions. A hierarchical variant additionally infers one of five force amplitude scales under a Matérn-$3/2$ time prior. This accounts for a force prior itself supplying apparent information about stiffness. Discretization, priors, and sensitivity to temporal mismatch are given in the supplement.

A view and a calibrated reset. For the spatial field study the target concatenates hidden displacements under the original loads and responses to four *new* known loads, with 192 scalar coordinates in each group. The design minimizes expected coordinate variance averaged over this fixed target. A new projected track on the original episode updates both the physical weights and conditional forces. A reset probe supplies a response to a known load and updates the physical weights alone. We allow four scalar measurements and approximate each one-step predictive expectation with 128 common-random-number draws. For the separate beam decision study, bending and base compliance replace the five spatial coordinates. A single tip observation constrains their sum, while either a view of the moving support or a calibrated push at another location can distinguish their contributions.

From video to dynamic response. The dynamic trials take calibrated cameras, a metric rest shape, and known load commands. The completed RGB-D stem study follows surface texture with pyramidal Lucas–Kanade tracking initialized from known rest points; noisy depth supplies the third coordinate. The RGB-only protocol instead follows each point in two overlapping fitting cameras and triangulates its position. Geometry and rest-point initialization are additional inputs to our estimator. GIC and PAC-NeRF reconstruct their object from video. The reserved camera participates only in evaluation.

For the stem, each candidate specifies E/ρ, ρ and Poisson's ratio; the support is clamped. A bank of coarse linear elastodynamic responses predicts motion under gravity and an applied force. With observed marker displacements y_i, predicted displacements $g_i(\theta)$ and observation noise σ, the

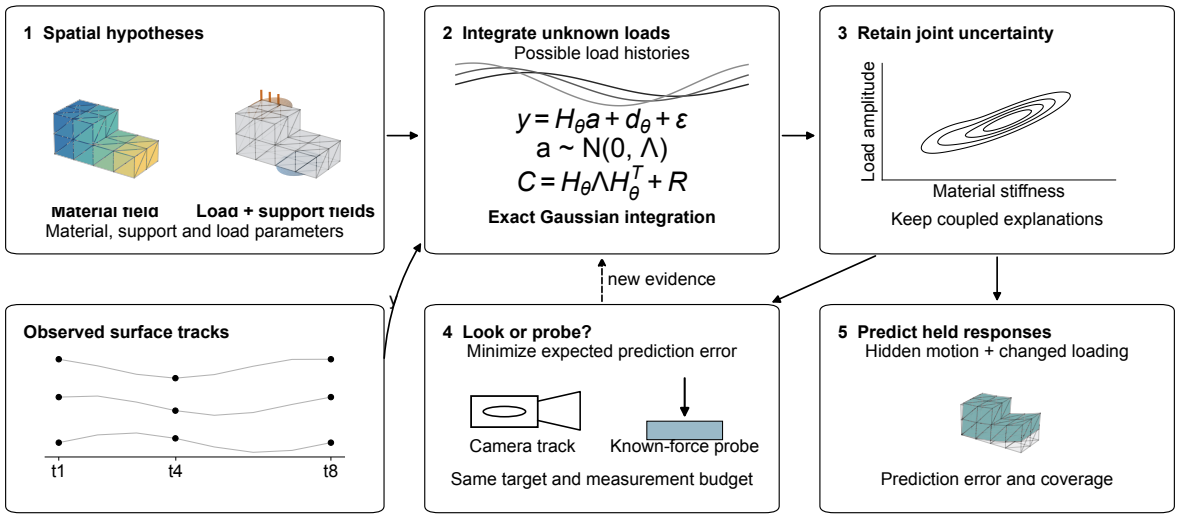


Figure 2. **The unknown-load branch of the method.** Each spatial hypothesis defines a response to unit forces. Integrating over force amplitudes assigns weights to material and support hypotheses and preserves their conditional force distributions. Candidate views and probes are compared in terms of the resulting change in predictive uncertainty. In the known-command dynamic experiments we use the measured command directly.

weights after each episode satisfy

$$w_\theta \propto p(\theta) \exp\left[-\frac{1}{2\sigma^2} \sum_i \|y_i - g_i(\theta)\|^2\right]. \qquad (10)$$

We keep the noise estimate from the initial fit for the calibration update. We apply the held-out command to each weighted hypothesis and compute its predictive mean and threshold risk. The RGB-only block protocol uses an analogous bank over inclusion position, stiffness contrast, background modulus and Poisson ratio, with a homogeneous block also permitted. Both published competitors use a single global modulus in the configured benchmark; a test of inference alone would require the same heterogeneous material family for all estimators. The striker protocol propagates the calibrated stem posterior through a compliant contact law and scores momentum transfer from the freely evolving sphere. The calibration command is fixed in these three scenarios. Selective acquisition is evaluated separately.

5. Experiments and Results

5.1. Does the observed motion identify the next response?

Fit and forecast on the same camera. We fitted the released GIC [3] and PAC-NeRF [7] implementations to eight frames of the synthetic Torus sequence from ten cameras. We reserved camera 0 throughout and measured mean per-frame foreground PSNR there. GIC's deformation network reproduced the observed motion at 20.47 dB, ahead of its physics replay at 17.79 dB. Over the next six frames the deformation network scored 8.29 dB, below its frozen last frame (9.50 dB), while the physics forecast reached 18.44 dB. PAC-NeRF's physics forecast scored 18.01 dB. This experiment tests continuation of observed motion. A separate intervention on three released ReconPhys assets [11] probes absolute scale. Doubling mass, stiffness and damping together gave 94.6–100.1 dB between simulated drop movies, yet the two versions responded differently to the same uniform force per anchor.

A calibrated intervention on visually identical stems. Three stems have densities of 500, 1000 and 2000 kg/m^3 and proportional elastic moduli of 2, 4 and 8 MPa. They sag identically under gravity. A 20 N lateral load at three quarters of the free length produces peak lateral displacements of 30.7, 15.4 and 7.7 mm in the nonlinear FEM reference. To test transfer, we sealed predictions of a new 25 N diagonal load at the same location before generating its reference. All methods received the known commands. GIC reconstructed the stem from ten RGB views and fixed density at 1000 kg/m^3. Our estimator used 16 surface points tracked in two RGB views with noisy rendered depth and known rest geometry. Its coarse linear tetrahedral model shares its element family with the nonlinear reference.

Calibration brought our errors below the 3 mm tolerance in all three cases (Table 1). In the earlier 20 N lateral-push trial, the same-bank posterior mode and posterior mean gave the same calibrated point forecast. GIC improved on the

Table 1. New diagonal push. RMS 3D displacement error (mm) across four stations and 37 frames, before and after one 40 N midpoint poke.

Density (kg/m^3)	500	1000	2000
GIC, RGB, before	14.33	3.08	7.18
GIC, RGB, after	12.98	1.88	8.08
Ours, RGB-D, before	8.74	5.30	12.08
Ours, RGB-D, after	2.75	1.12	0.65

two lighter stems and crossed the threshold at its assumed density. On the withheld camera its deformation network rendered passive sag at 24.9 dB and its physics replay at 20.6 dB. For the light stem, the passive physics forecast scored 16.3 dB on the diagonal push. A subsequent rendering analysis (Fig. 3) drove the same fixed GIC appearance with our calibrated motion or the reference motion, yielding 21.27 and 21.49 dB, respectively. We sealed the motion forecasts prospectively and finalized the common-renderer analysis and its foreground-error aggregation separately.

Choosing whether to measure. In an independent beam study, 96 draws of bending and base compliance each faced three task or visibility conditions. All policies saw the same noisy tip measurement. With a tolerance of 0.50 mm, a view and a calibrated push cost 0.015 and 0.040 in failure-probability units. For a repeated tip load the risk policy proceeded in all 96 cases (0.104 mm RMSE). For a shifted load the initial RMSE was 0.939 mm. With a visible base it acquired a view in 92 cases (0.073 mm). With a concealed base it probed in 91 (0.119 mm). It proceeded in the remaining cases. Against a cost-aware rule using Gaussian moments, the paired decision-loss difference (full minus Gaussian) was -0.00033 with a 95% world-bootstrap interval of $[-0.00085, 0]$.

Image and impulse rankings under contact. In an earlier six-family study, adapted Spring-Gaus [14] and PhysTwin [6] mechanics received identical calibration tracks, geometry, density and boundary conditions. A common renderer displayed their forecasts. On eight held-out striker impacts across four material worlds, Spring-Gaus exceeded PhysTwin's foreground PSNR in every world (mean advantage 1.22 dB), yet its 3D motion error was higher (4.80 versus 4.01 mm) and its net impulse error was 0.0419 versus 0.0231 N s. On eight curved-obstacle pushes Spring-Gaus gained 2.35 dB and reduced motion error from 4.97 to 3.15 mm. These measurements characterize the two adapted mechanics implementations under supplied tracks.

Sensitivity to model error. A deformable body in a concave bowl was pushed or tilted once before a stronger transverse push. Across 16 material/contact settings with two noise draws each, the coarse model predicted 1.86% failure after adaptive calibration, compared with 43.75% realized against a finer simulation. A mixture including the finer solver by construction improved coverage. Risk estimates consequently depend on which mechanical discrepancies the posterior represents.

Table 2. **Prediction after four scalar observations.** Means over 128 matched cases. Hidden and new-load errors are coordinate RMSE in mm. Material error is relative; force error is in newtons. Fixed denotes the development-selected mixed schedule.

Policy	Hidden mm	New load mm	Material rel.	Force N
Initial	0.076	0.024	0.451	0.042
Active	0.045	0.014	0.347	0.030
Fixed	0.057	0.012	0.322	0.033
Fixed View	0.050	0.022	0.421	0.032
Fixed Probe	0.074	0.010	0.277	0.040
Random	0.059	0.014	0.343	0.034

Three-scenario comparison protocol. The RGB-only protocol pairs GIC, PAC-NeRF and our estimator on a new stem push, an off-centre press on a block with a hidden stiff region, and a freely rebounding striker. The push and impact reuse the stem geometry. The protocol specifies triangulated RGB observations and held-out displacement or impulse. Aggregate results for this comparison are pending.

5.2. Joint spatial identification under unknown loading

We also evaluate when a view or a known force provides better evidence in the presence of unknown loading. We use a block and stepped block of bounding dimensions $120 \times 60 \times 60$ mm. Inference starts from 24 scalar observations of projected motion on the visible surface: four material points, two orthographic components and three loading times. Additional tracks use an oblique camera; a geometric visibility check covers all deformed test states. Sensor noise is 0.02 mm. The force standard deviation and calibrated probe amplitude are 0.08 N. The two geometry-specific development sets select fixed schedules before held-out evaluation. The supplement describes the discretization and all independent cohorts.

Acquisition and the target. Across 128 matched cases, four adaptive measurements reduce combined coordinate RMSE from 0.0578 to 0.0340 mm. Fixed camera-only, fixed mixed and random policies reach 0.0403, 0.0423 and 0.0438 mm. Against the fixed mixed schedule, the paired difference is -0.0083 mm (95% case-bootstrap in-

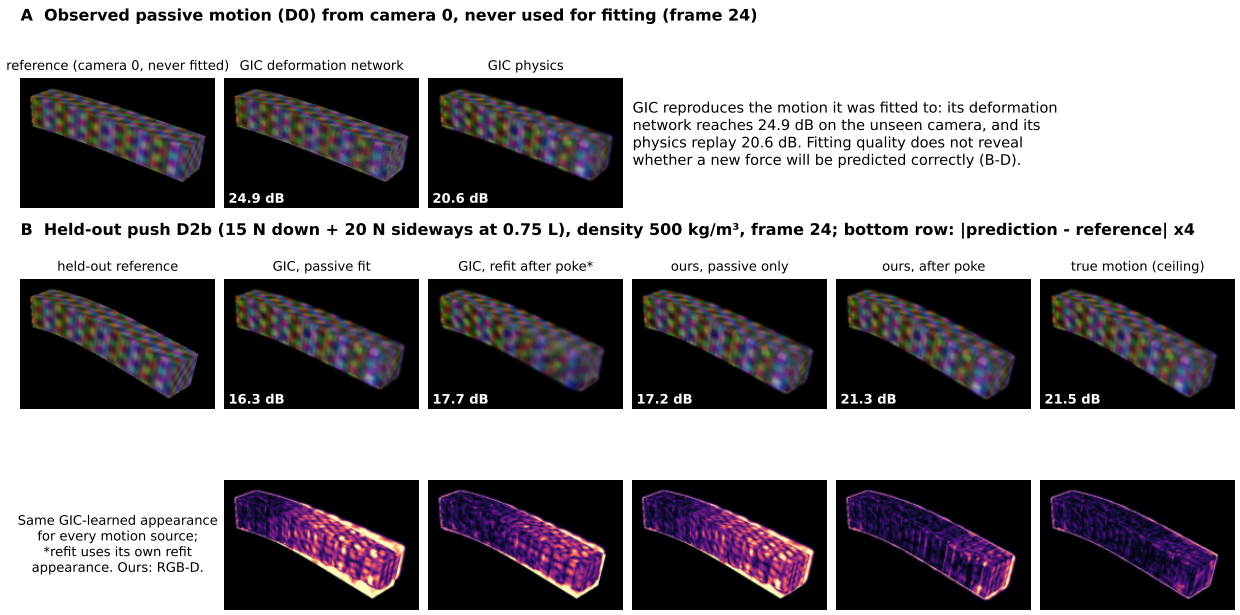


Figure 3. **Rendering the measured and predicted stem motion from a withheld camera.** The upper row shows a passive frame. The lower panels show the new diagonal push on the light stem with pixel differences magnified fourfold. Numbers in the images are sequence foreground PSNR. Comparisons primarily use one frozen GIC appearance; the GIC refit uses its own appearance. The motion predictions were sealed before reference generation; the common-appearance analysis was finalized subsequently.

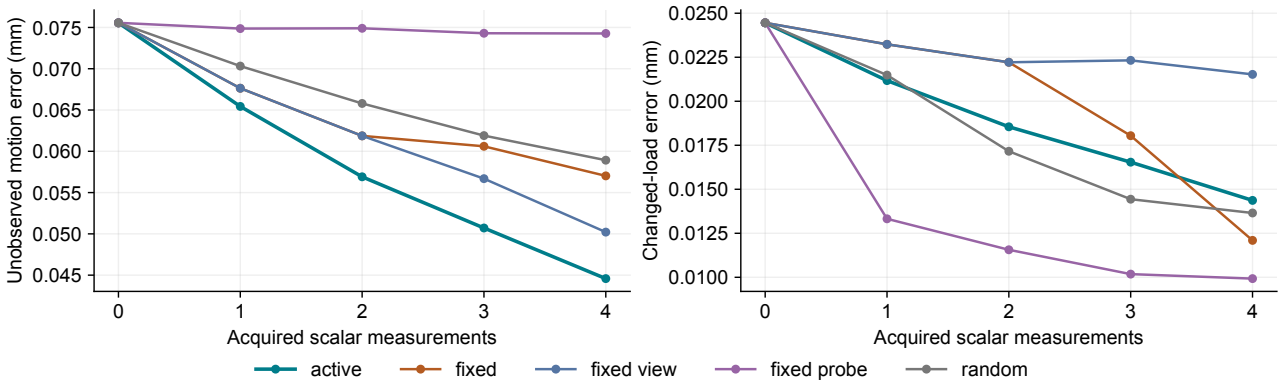


Figure 4. **Measurement choice depends on the target.** An additional camera can expose the original hidden motion, whereas a known probe directly constrains response under a changed load. Curves average all matched surface-track cases.

terval $[-0.0110, -0.0058]$). The adaptive policy spends 439 of its 512 measurements on camera observations. Probes alone give the best prediction under known new loads, at 0.0099 mm against 0.0144 mm for adaptive selection, but leave more uncertainty in the original hidden motion (Table 2, Fig. 4). In another 128 cases with uncertain force scale, four fixed views reduce hidden-motion error by 33.4%, while four calibrated probes reduce new-load error by 48.4%. Both branches start with the same evidence.

What retaining dependence changes. Retaining the conditional force distribution for each material/support hypothesis gives a mean prediction-interval width of 0.0717 mm; replacing those conditionals with a common moment-matched distribution gives 0.0863 mm. Coverage is 92.0% versus 93.9% at nominal 90%, and hidden-motion errors are nearly equal (0.0446 and 0.0449 mm). In this experiment the practical gain is narrower intervals at similar accuracy. On a separate continuous-prior cohort, sequential Monte Carlo gives 0.0294 mm after the recorded adaptive actions versus 0.0350 mm after fixed mixed observations.

Calibration outside the fitted family. Finite-model prediction intervals cover 92.0% of matched target coordinates. Coverage falls to 86.4% for continuous parameters, 86.1% when the generator uses a finer mesh, and 79.2% for a shorter correlation length in the true force history. A follow-up that mixed the temporal length and amplitude on 64 fresh stress cases improved force coverage from 72.4% to 83.9%; inferred material error increased. More observations can narrow a posterior around the wrong explanation.

6. Discussion

Visual evidence is informative about the motion it reveals. A change in load, support or contact tests which mechanical explanations remain compatible with the response a robot needs. The observed rendering scores, the calibrated-force trials and the choice to look or probe address different aspects of that question. A camera change can disclose support movement; even many extra views of an identical 3D sag leave the force scale ambiguous. For the diagonal push, the GIC simulator supplied with the true material parameters still has 7.1–9.3 mm station error. Its reconstructed shape, fixed-density identification and MPM dynamics all affect the comparison. Our estimator knows the rest shape, receives depth in the completed prospective trial, and uses the same element family as the finite simulation reference. These numbers establish a controlled counterexample and a repeatable calibration effect under the stated sensing conditions. Ranking full systems on real RGB footage requires a matched-input test.

Accurate risk predictions require a posterior broad enough to represent sensor and simulator error. The bowl result quantifies the cost of ignoring this condition. Similarly, the beam policy's small advantage over a cost-aware Gaussian approximation has an interval including zero. A real-object transfer experiment would need measured inputs for calibration, an independently observed response to a different interaction, and evaluation of failures across objects. Known geometry, track initialization and the single-step cost model remain practical restrictions. In the present RGB protocol, the physical likelihood uses tracked motion; image rendering provides an independent visual readout. Weighting hypotheses directly by visible silhouettes or reliable multiview pixel residuals is a concrete extension to test on fresh episodes.

7. Conclusion

An observation identifies a response only to the extent that compatible models agree about the intended interaction. Propagating joint physical uncertainty through that interaction gives a testable prediction and a reason to seek specific evidence. Our experiments show when another view or a known probe resolves the ambiguity, and how a misspecified simulator can make that decision overconfident.

LLM Usage Statement

OpenAI ChatGPT and coding agents, and Anthropic Claude, assisted with implementation, synthetic experiment orchestration, numerical and plotting scripts, literature organization, and manuscript preparation. ChatGPT was also used for reviewer-style feedback, exploration of experimental hypotheses, and editing and drafting text. Numerical figures were produced by scripts from simulation outputs or recorded scores. The authors take responsibility for the submitted text, references, code, and results.

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